COMPETITIVE PROGRAMMING PROBLEMS

1. Find Duplicates · O(n²) Time, O(1) Space

```
#include <stdio.h>

int main() {

int n;

scanf("%d", &n);

int count[n + 1];

for (int i = 0; i <= n; i++)

count[i] = 0;

for (int i = 0; i < n; i++) {

int x;

scanf("%d", &x);

count[x]++;

if (count[x] == 2) {

printf("%d", x);

return 0;

}

}

return 0;
```

```
}
```

2. Find Duplicates · O(n) Time, O(1) Space

```
#include <stdio.h>

int main() {

int n;

scanf("%d", &n);

int a[n];

for (int i = 0; i < n; i++)

scanf("%d", &a[i]);

for (int i = 0; i < n; i++) {

for (int j = i + 1; j < n; j++) {

if (a[i] == a[j]) {

printf("%d",

a[i]); return 0;

}

}

}

return 0;
}
```

3. Intersection of Two Sorted Arrays · O(m × n) Time, O(1) Space

```
#include <stdio.h>

int main() {

int T;

scanf("%d", &T);

while (T--) {

int n1, n2;

scanf("%d", &n1);

int a[n1];

for (int i = 0; i < n1; i++)

scanf("%d", &a[i]);

scanf("%d", &n2);

int b[n2];

for (int i = 0; i < n2; i++)

scanf("%d", &b[i]);

int i = 0, j = 0;

int first = 1;

while (i < n1 && j < n2) {
if (a[i] == b[j]) {
```

```
if (!first)

printf(" ");

printf("%d",

a[i]); first = 0;

i++;

j++;

}

else if (a[i] < b[j]) {

i++;

}

else {

j++;

}

}

printf("\n");

}

return 0;

}
```